EXPLORATORY DATA ANALYSIS

Chapter 24

Building a Professional EDA Report

Turning analysis into a clear, trustworthy, decision-focused deliverable.

Python • NumPy • Pandas • Matplotlib (object-oriented API) • Seaborn

A professional report does more than display charts. It explains what was investigated, what the data can support, what was found, and what remains uncertain.

1. Learning Objectives

- Design an EDA report around the reader's questions and decisions.
- Write an executive summary that communicates the main findings without exaggeration.
- Describe the dataset, observational unit, scope, and methodology clearly.
- Present data-quality findings, quantitative summaries, and visual evidence.
- Choose tables and figures that answer specific analytical questions.
- Write captions and interpretations that include context and denominators.
- Explain limitations, uncertainty, and recommendations responsibly.
- Build a repeatable report-generation workflow from saved EDA outputs.

2. Review of Earlier Chapters

This review revisits the foundations needed to communicate EDA results. Chapter 23, the immediately previous chapter, is reviewed separately in Section 3.

- Chapters 1–3: EDA is investigation, not automatic explanation. Understand the observational unit, metadata, and data quality before drawing conclusions.
- Chapters 4–9: numerical and categorical summaries, time variables, distributions, quantiles, skewness, and kurtosis need precise definitions and denominators.
- Chapters 10–12: visual encodings, scales, bins, facets, and plot selection shape interpretation. A chart must fit the variable and question.
- Chapters 13–15: association, covariance, dependence, and grouped analysis require clear group definitions, sample sizes, and careful interpretation.
- Chapters 16–17: unusual observations require investigation and context; detection does not justify automatic deletion.
- Chapter 18: high-dimensional results need careful selection, multiplicity awareness, and readable summaries.
- Chapter 19: time-series reports must state timezone, cadence, aggregation, and temporal coverage.
- Chapter 20: sampling and data-collection processes constrain generalization.
- Chapter 21: automated profiling is a screening aid, not a substitute for interpretation.
- Chapter 22: domain context determines the meaning of rows, variables, risks, and denominators.

3. Review of Chapter 23 — Designing a Reproducible EDA Workflow

Chapter 23 focused on making analysis inspectable and rerunnable: preserving raw data, recording provenance, centralizing configuration, validating inputs, managing dependencies, structuring code, running notebooks from a clean kernel, saving outputs, logging, and using Git responsibly.

Expected reasoning for the previous assignment/project: a sound project should make it possible to identify the input dataset and version, understand the analysis settings, rerun the notebook, locate generated tables and figures, and trace important exclusions or decisions. It should also protect private data and secrets.

A professional report depends on this foundation. If the source data, filters, or definitions are unclear, polished formatting cannot make the findings trustworthy.

4. What Is a Professional EDA Report?

An EDA report is a structured explanation of an investigation into a dataset. It documents the scope, data, methods, evidence, findings, limitations, and useful next steps.

It is not necessarily a research paper, dashboard, or machine-learning model report. Its purpose is to help a reader understand the dataset and the patterns or risks discovered during exploration.

4.1 The report should answer five questions

- **Why?** What question or business problem motivated the analysis?
- **What?** What data was examined, and what does one row represent?
- **How?** What checks, summaries, comparisons, and visual methods were used?
- **What was found?** Which observations are supported by the evidence?
- **So what?** What should the reader consider next, and what cannot yet be concluded?

4.2 Report quality is not decoration

A report can look beautiful and still be analytically weak. Professional quality comes from correct analysis, transparent definitions, relevant evidence, readable presentation, and honest limitations. Design should make reasoning easier to follow.

5. Know the Reader and the Decision

Before writing, identify the intended reader. A business owner, product manager, researcher, and technical analyst may need different levels of detail.

- What decision or question brought the reader to this report?
- How familiar is the reader with statistics and the data domain?
- Which findings are immediately useful?
- What evidence does the reader need to trust each finding?
- Which technical details belong in the main report and which belong in an appendix?

For a non-technical reader, explain the meaning of a statistic rather than assuming the reader knows its name. For a technical reader, include definitions, sample sizes, methods, and reproducibility details.

5.1 Separate the question from the method

Weak framing: 'I made a correlation heatmap.' Strong framing: 'I examined whether order value moves with quantity and customer tenure; the correlation summary is one tool used to investigate that question.' Start with the question, then explain why a method is appropriate.

6. Recommended Report Structure

The exact structure depends on the project, but the following is a strong default for a client-facing EDA report.

```
1. Title page
2. Executive summary
3. Objectives and key questions
4. Dataset overview and scope
5. Data quality and preparation notes
6. Methodology
7. Main findings
   7.1 Univariate patterns
   7.2 Group differences
   7.3 Relationships and associations
   7.4 Time or domain-specific patterns
   7.5 Unusual observations
8. Limitations and uncertainty
9. Conclusions and recommended next steps
10. Appendix
    - Detailed tables
    - Additional figures
    - Definitions and technical notes
    - Reproducibility information
```

Do not force every section into every project. A small dataset may need a short report. A complex project may need appendices or a separate technical document.

7. Executive Summary

The executive summary is often the most-read part of the report. Write it after the analysis is complete, even though it appears near the beginning.

7.1 What to include

- Purpose: what the analysis investigated.
- Scope: the dataset, period, and population or sample examined.
- Two to five important findings, expressed in plain language.
- Important data-quality caveats that affect interpretation.
- A small number of practical next steps.

7.2 Example

Purpose: This analysis explored order patterns in the supplied online retail dataset for January–December 2025.

Scope: The dataset contained 48,200 transaction-line records. One row represented a product line within an order, not a unique customer.

Findings: Order volume was concentrated in a small number of product categories. Median line value was lower than the mean, indicating a right-skewed distribution. Monthly totals varied, with a noticeable increase during the final quarter.

Caveat: The data did not include marketing exposure or inventory availability, so the analysis cannot explain why the observed patterns occurred.

Next step: Validate the final-quarter increase against campaign, pricing, and stock records before using it for planning.

These figures are illustrative examples, not findings from your own dataset. Never copy sample numbers into a real client report.

7.3 Avoid these executive-summary mistakes

- Listing every chart instead of communicating the most important findings.
- Using unexplained statistical terms or abbreviations.
- Claiming causes from descriptive patterns.
- Hiding major data-quality problems in the appendix.
- Making recommendations that are not connected to the evidence.

8. Dataset Overview and Scope

A reader must understand what the dataset represents before interpreting results.

8.1 Essential dataset description

- Source and how the data was obtained, subject to confidentiality.
- Time period and date of extraction.
- Number of rows and columns, including any important changes during preparation.
- Observational unit: what exactly does one row represent?
- Population or sample represented, if known.
- Important identifiers, variables, units, and categories.
- Inclusion and exclusion criteria.
- Known gaps in coverage or collection.

8.2 Define the analytical population

The raw dataset and the population used for a particular chart may differ. For example, a revenue analysis may exclude cancelled orders, while an operational analysis may include them. State the filters and counts for each important analysis.

Do not call a dataset a random sample unless the sampling design supports that statement. A convenient export or voluntary survey may not represent the broader population.

9. Data Quality and Preparation Notes

Data quality findings belong in the report when they affect trust, interpretation, or the scope of the analysis. Do not bury important problems.

9.1 Useful quality dimensions

- Completeness: missing values and missing records.
- Validity: values that violate known formats or domain rules.
- Consistency: conflicting representations or incompatible values.
- Uniqueness: duplicate records and identifier problems.
- Timeliness: whether the data covers the required period and is current enough.
- Accuracy: whether values agree with a trusted reference, when one exists.

9.2 Report quantities with denominators

Instead of writing 'There were 350 missing values', say which field was affected, how many records were eligible, and the percentage missing.

```
missing_summary = (
    df.isna()
      .sum()
      .rename("missing_count")
      .to_frame()
)

missing_summary["row_count"] = len(df)
missing_summary["missing_rate"] = (
    missing_summary["missing_count"] / len(df)
)

missing_summary["missing_percent"] = (
    100 * missing_summary["missing_rate"]
)
```

A column-level missing percentage uses the number of rows as its denominator. If the analysis uses a filtered subset, calculate and report the relevant denominator for that subset.

9.3 Explain preparation without hiding it

Document parsing, type corrections, deduplication, exclusions, and derived variables. Distinguish source-data corrections from analytical filters. If a row was excluded, state the rule and quantify the impact.

10. Methodology: Explain What You Did

The methodology section should be detailed enough for a reader to understand how the findings were produced, without becoming a copy of the entire notebook.

- Define the variables and units used in important analyses.
- State filters, date ranges, and group definitions.
- Explain how missing values were handled for each relevant calculation.
- Name important statistics and association measures.
- Describe plot types and any important scale or bin choices.
- Explain how unusual observations were flagged and investigated.
- Mention software and package versions when useful for reproducibility.

Avoid vague phrases such as 'standard statistical methods were used'. If a method influences interpretation, name it and explain the relevant choice.

10.1 Method choice should fit the question

- Use median and IQR when a numerical distribution is strongly skewed or contains influential extremes.
- Use a histogram or ECDF to examine distribution shape; state binning choices when they matter.
- Use a grouped summary when the question compares meaningful categories or cohorts.
- Use Pearson correlation for linear association between numerical variables when its limitations are acceptable; consider rank-based measures for monotonic relationships or influential outliers.
- Use a contingency table and an appropriate association measure for categorical variables.
- Use time-aware summaries for temporal data; do not treat irregular events as equally spaced observations without justification.

11. Presenting Quantitative Findings

Numbers are evidence, but numbers without context are difficult to interpret.

11.1 Give the reader a useful comparison

Weak: 'The median amount was 420.' Stronger: 'The median transaction-line amount was 420 currency units across 12,450 valid records; the mean was 610, reflecting a right-skewed distribution.'

The stronger statement identifies the variable, unit, sample size, comparison, and interpretation.

11.2 Choose precision deliberately

- Avoid displaying many decimal places when they do not change the interpretation.
- Use consistent units and rounding throughout the report.
- Do not round intermediate calculations if that could affect final results.
- Label percentages and currency units explicitly.
- Do not compare percentages with different denominators as if they were directly comparable.

11.3 Counts, rates, and weighted averages

A larger group may have a larger count simply because it contains more observations. When comparing groups, consider whether counts, proportions, per-unit rates, or weighted averages answer the question. State the denominator and weighting basis.

12. Tables That Communicate

Tables are useful when readers need exact values, multiple metrics, or detailed group comparisons.

12.1 Table design principles

- Give every table a descriptive title and a number or identifier if the report is long.
- Use meaningful row and column labels, units, and consistent rounding.
- Sort categories by a meaningful order, not automatically alphabetically.
- Include sample sizes and denominators where needed.
- Avoid excessive columns and repeated information.
- Highlight only values that support the report's purpose.
- Explain abbreviations and special missing-value codes.

12.2 Example: grouped summary

Illustrative structure:

```
| Category | Records | Median amount | Mean amount |
|---|---:|---:|---:|
| Books | 1,250 | 18.00 | 22.40 |
| Clothing | 2,100 | 35.00 | 48.10 |
| Electronics | 980 | 72.00 | 96.30 |
```

A table like this still needs context: What does one record represent? Are cancelled orders included? Are the amounts in USD, BDT, or another currency? Are missing values excluded from the summaries?

13. Figures That Communicate

A figure should answer a question, not merely fill space.

13.1 Choose the visual by analytical purpose

- Histogram or ECDF: What does a numerical distribution look like?
- Boxplot or violin plot: How do distributions differ across groups?
- Bar chart: How frequent are categories or events?
- Scatterplot: How do two numerical variables relate?
- Heatmap: How does a matrix of values or associations vary?
- Line chart: How does a measure change over time?
- Small multiples: How does the same pattern vary across meaningful groups?

13.2 Figure checklist

- Descriptive title or clear caption.
- Axes labelled with variable names and units.
- Readable tick labels and legend, if needed.
- Appropriate scale and range.
- Sample size or relevant denominator.
- Time window and filters, when relevant.
- Source or dataset reference, where appropriate.
- No misleading truncation, unnecessary 3D effects, or decorative clutter.

13.3 Use captions to explain the evidence

A useful caption tells the reader what is shown, which observations are included, and how to interpret the important feature.

Weak caption: 'Sales by month.'

Better caption: 'Monthly completed-order revenue, January–December 2025. Cancelled orders are excluded; months are shown in UTC. The increase in November is descriptive and does not identify its cause.'

13.4 Matplotlib object-oriented example

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(8, 4.5))

ax.plot(
    monthly_summary["month"],
    monthly_summary["revenue"],
    marker="o"
)

ax.set(
    title="Monthly completed-order revenue",
    xlabel="Month",
    ylabel="Revenue (BDT)"
)
ax.grid(axis="y", alpha=0.25)
fig.tight_layout()
fig.savefig(
    "outputs/figures/monthly_revenue.png",
    dpi=180,
    bbox_inches="tight"
)
plt.close(fig)
```

This example assumes monthly_summary has already been computed with a documented definition of completed orders and revenue. The plot code alone does not define the analytical population.

14. Writing Findings: Observation, Interpretation, Explanation

One of the most important professional skills is separating what the data shows from what you think may explain it.

14.1 Three levels of a finding

- **Observation:** what is directly visible or calculated from the data.
- **Interpretation:** what the pattern may mean within the known context.
- **Explanation:** a proposed reason or mechanism, which may require additional evidence.

Example

Observation: Orders in the final quarter were higher than in the previous quarter.

Interpretation: The dataset shows a seasonal or period-specific increase in recorded order volume.

Possible explanation: Promotions, holidays, price changes, or improved stock availability may have contributed. The current dataset does not establish which explanation is correct.

This distinction prevents descriptive EDA from turning into unsupported causal claims.

14.2 A practical finding template

Finding: [What was observed, with a number and denominator.]

Evidence: [Which table, statistic, or figure supports it.]

Interpretation: [What the pattern means in context.]

Caveat: [What the evidence does not establish.]

Next step: [What additional check or action is reasonable.]

15. Limitations and Analytical Uncertainty

Every dataset has limits. A professional report makes important limitations visible rather than pretending the analysis is complete.

- Coverage: which people, events, locations, or time periods are absent?
- Selection: how were records included, and who may be missing?
- Measurement: are variables proxies, self-reports, estimates, or system-generated values?
- Missingness: could missing values be systematic?
- Granularity: does the row-level structure support the question being asked?
- Confounding: could other variables explain an observed association?
- Time: could collection changes, seasonality, or structural breaks affect comparisons?
- Uncertainty: how stable are the patterns, and how sensitive are they to definitions or exclusions?

Do not use a generic disclaimer as a substitute for explaining the actual limitations of the dataset.

15.1 Sensitivity checks

When a conclusion depends on a choice, compare reasonable alternatives. Examples include using mean versus median, including versus excluding a flagged but valid extreme, comparing multiple time windows, or reporting results with and without a clearly defined subset. Explain whether the main conclusion changes.

15.2 Avoid false certainty

EDA is usually descriptive. A strong pattern is not automatically statistically significant, practically important, representative, or causal. Use inferential statistics only when the design and question justify them, and explain the assumptions.

16. Conclusions and Recommendations

The conclusion should synthesize the most important findings, not repeat every chart.

16.1 Recommendations must follow from evidence

- Connect each recommendation to a finding or a clearly stated information gap.
- Distinguish an action supported now from an action that needs further validation.
- Use practical language and identify what additional data would help.
- Avoid prescribing a business decision outside the scope of the analysis.
- Prioritize a small number of useful next steps.

Example: 'Before changing inventory targets, compare the observed category-level demand pattern with stockout records. The current order data measures completed transactions, not unmet demand.'

16.2 What EDA cannot promise

EDA cannot guarantee future performance, prove causation from observational patterns alone, or establish population-wide behavior when the sample design does not support it. State what the analysis can support and what requires another study or data source.

17. Appendix and Technical Detail

The appendix keeps the main report readable while preserving useful detail.

- Full variable dictionary or selected variable definitions.
- Detailed quality-audit tables.
- Additional distributions and subgroup plots.
- Technical method notes and formulas.
- Filter and exclusion definitions.
- Reproducibility information: source version, execution date, software versions, and project location, where safe to share.
- Additional checks that support but do not belong in the main narrative.

Do not use the appendix as a dumping ground. Include material that helps a reader verify or extend the analysis.

18. Report Production Workflow

A practical workflow for producing a report:

1. Confirm the audience, purpose, and questions.
2. Review the reproducible project and verify the dataset version.
3. Select findings that answer the questions; do not include every output.
4. Prepare tables with definitions, denominators, and consistent precision.
5. Prepare figures with clear titles, units, scales, and captions.
6. Draft findings using observation, interpretation, caveat, and next step.
7. Write limitations and check every recommendation against evidence.
8. Write the executive summary last.
9. Run a consistency check: numbers in prose must match tables and figures.

10. Review readability, privacy, file names, and delivery completeness.

A report is not finished when the PDF exports successfully. It is finished when its claims have been checked against the analysis and its intended reader can understand the message.

19. Common Reporting Mistakes

- Starting with charts before defining the question and audience.
- Reporting a statistic without a unit, denominator, or sample size.
- Using too many figures without a narrative.
- Repeating the same information in text, tables, and charts.
- Choosing a chart because it looks attractive rather than because it answers the question.
- Presenting correlation as causation.
- Hiding missingness, exclusions, or sample limitations.
- Overstating small differences or uncertain patterns.
- Making recommendations that are not supported by the analysis.
- Using inconsistent terminology, units, rounding, or date formats.
- Writing an executive summary that is only a list of methods.
- Including private or identifying information in a report or its metadata.

20. Summary

- A professional EDA report is a structured explanation of questions, data, methods, evidence, findings, and limitations.
- Write for a specific reader and decision.
- Define the dataset, observational unit, scope, and analytical population.
- Report data quality transparently and use meaningful denominators.
- Use tables for exact comparisons and figures for patterns; give both enough context.
- Separate observation, interpretation, and possible explanation.
- State limitations and uncertainty honestly.
- Make recommendations that follow from evidence.
- Use appendices for useful technical detail and verify all report values before delivery.

21. Assignment

Part A — Concepts

- What is the difference between an EDA report and a collection of charts?
- Why should the executive summary be written after the analysis?
- Explain the difference between an observation, interpretation, and explanation.
- Why must percentages include a denominator?
- Give three limitations that could affect the interpretation of an online retail dataset.
- When should technical details be placed in an appendix?

Part B — Rewrite findings

Rewrite each statement professionally, adding context or a caveat where necessary:

- 'Sales are better in November because of marketing.'

- 'There are lots of missing values.'
- 'Category A is the best category.'
- 'The two variables are correlated, so one affects the other.'

Part C — Report outline

Using an EDA dataset from an earlier chapter, prepare a report outline with the audience, purpose, dataset scope, five key questions, planned tables/figures, and likely limitations. Do not invent findings before examining the evidence.

Part D — Critical review

Imagine a report shows that one customer group has twice the total revenue of another. List at least four checks you would perform before recommending that the business prioritize the larger-revenue group.

22. Coding Project — Client-Ready EDA Report

Objective: produce a concise, professional report from the reproducible EDA project developed in Chapter 23.

Requirements

- Choose a clear audience and write a one-paragraph purpose statement.
- Describe the dataset, observational unit, time period, and analytical population.
- Include a data-quality section with counts and denominators.
- Include at least three quantitative findings supported by reproducible calculations.
- Include at least four well-chosen figures, each with labels, units, and useful captions.
- Include at least one meaningful group comparison and one relationship analysis where the data supports them.
- For every major finding, distinguish observation from interpretation and possible explanation.
- Document filters, exclusions, and treatment of missing or unusual observations.
- Include limitations and evidence-linked next steps.
- Add an appendix with technical notes and reproducibility information that is safe to share.
- Verify all numbers in the report against the saved tables and outputs.

Deliverables: PDF report, source notebook, reusable code, selected tables and figures, README or methodology note, and a short list of limitations.

Master notebook instruction: continue extending your single cumulative EDA master notebook. Use the reproducible outputs and analysis patterns already developed; do not create a disconnected permanent notebook for this chapter.

Evaluation criteria: analytical correctness, clarity, audience fit, quality of evidence, figure/table design, transparency, responsible interpretation, and consistency between report and code.

23. Preview of Next Chapter

Chapter 25 — Communicating Insights and Analytical Uncertainty

The final chapter will focus on communicating findings clearly in conversation, presentations, and client discussions. We will examine how to prioritize insights, explain uncertainty without confusing the audience, answer challenging questions, avoid misleading narratives, and communicate what the analysis does—and does not—show.

Chapter 24 builds the report. Chapter 25 develops the communication skills needed to make its insights understood and used.